



DW_fp_exp

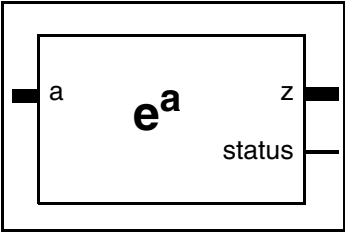
Floating-Point Exponential (e^a)

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Features and Benefits

- The precision is controlled by parameters, and covers formats in the IEEE Standard 754
- Exponents can range from 3 to 31 bits
- Fractional part of the floating-point number can range from 2 to 57 bits
- A parameter controls the use of denormal values.

Revision History



Description

DW_fp_exp computes the exponential of a floating-point input *a*, delivering an output $z = e^a$, which is also a floating-point value.

A list of all the parameters used to configure this component is shown in [Table 1-2](#).

The parameter *ieee_compliance* controls the use of denormals and NaNs, as done for other FP operators in the DesignWare Library. When *ieee_compliance* = 0, the operator takes NaN values as infinities, and denormals as zeros. When *ieee_compliance* = 1, the component accepts and generates denormalized values, handles NaN inputs, and delivers NaN outputs when necessary.

Parameters *sig_width* (significand field size) and *exp_width* (exponent field size) define the floating-point format used by input and output operands. Some of these floating-point formats match the formats defined in the IEEE Standard 754.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	(<i>sig_width</i> + <i>exp_width</i> + 1) bits	Input	Input data
z	(<i>sig_width</i> + <i>exp_width</i> + 1) bits	Output	Exponential = e^a
status	8 bits	Output	Status flags for the result For details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> .

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 57 ^a	Word length of fraction field of floating-point numbers <i>a</i> and <i>z</i>
exp_width	3 to 31	Word length of biased exponent of floating-point numbers <i>a</i> and <i>z</i>

Table 1-2 Parameter Description (Continued)

Parameter	Values	Description
ieee_compliance	0 or 1 Default: 0	Controls the use of denormals and NaNs 0 = Do not use denormals or NaNs 1 = Use denormals and NaNs
arch	0 to 2 Default: 2	Implementation selection 0 = Area optimized 1 = Speed optimized 2 = Obsolete implementation retained for backward compatibility; will be removed in a subsequent release

a. The synthesis model fully supports this range, as does the Verilog simulation model in VCS, but the VHDL simulation model (in all simulators) and the Verilog simulation model in non-VCS simulators are limited to a range of 2 - 35.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Implement using the Datapath Generator technology combined with static DesignWare components	DesignWare

Table 1-4 Simulation Model

Model	Function
DW02.DW_FP_EXP_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_exp_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_exp.v	Verilog simulation model source code

Given the properties of algorithms to compute the exponential function, and the goal to have a component with good QoR, this component does not have rounding mode control as other FP components in the library. The error is bounded to have a maximum of 2 ulps.

The *arch* parameter controls implementation alternatives for this component. Different values result in different numerical behavior, but the error on the computed values is always bounded by a maximum of 2 ulps. You should experiment with this parameter to find out which value provides the best QoR for your design constraints and technology. Using *arch* = 0 (area optimized implementation) usually provides the best QoR for most time constraints.

For information about the floating-point system defined for the floating-point components, including status flag bits and floating-point formats, refer to the [Datapath Floating-Point Overview](#).

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_exp_inst is
    generic (
        inst_sig_width : POSITIVE := 10;
        inst_exp_width  : POSITIVE := 5;
        inst_ieee_compliance : INTEGER := 0;
        inst_arch       : INTEGER := 2
    );
    port (
        inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        status_inst : out std_logic_vector(7 downto 0)
    );
end DW_fp_exp_inst;

architecture inst of DW_fp_exp_inst is

begin

    -- Instance of DW_fp_exp
    U1 : DW_fp_exp
    generic map (
        sig_width => inst_sig_width,
        exp_width  => inst_exp_width,
        ieee_compliance => inst_ieee_compliance,
        arch => inst_arch
    )
    port map (
        a => inst_a,
        z => z_inst,
        status => status_inst
    );

end inst;

-- pragma translate_off
configuration DW_fp_exp_cfg_inst of DW_fp_exp_inst is
    for inst
    end for; -- inst
end DW_fp_exp_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_exp_inst( inst_a, z_inst, status_inst );

parameter inst_sig_width = 10;
parameter inst_exp_width = 5;
parameter inst_ieee_compliance = 0;
parameter inst_arch = 2;

input [inst_sig_width+inst_exp_width : 0] inst_a;
output [inst_sig_width+inst_exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_fp_exp
DW_fp_exp #(inst_sig_width, inst_exp_width, inst_ieee_compliance, inst_arch) U1 (
    .a(inst_a),
    .z(z_inst),
    .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
March 2024	DWBB_202312.2	Fixed a link to the <i>Datapath Floating-Point Overview</i> on page 2
December 2023	DWBB_202312.0	Updated date and version; no technical changes
October 2022	DWBB_202203.5	Clarified error bounding in ulps on page 2
March 2019	DWBB_201903.0	Clarified value '2' of the arch parameter in Table 1-2 on page 1; this value will be obsoleted in a subsequent release
July 2018	DWBB_201806.1	- For STAR 9001366624, in Table 1-2 on page 1, clarified the range of <i>sig_width</i> for the VHDL simulation model (in all simulators) and the Verilog simulation model for non-VCS simulators. - Added this Revision History table and the document links on this page

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